INCH-POUND

MIL-L-51501A(EA)
11 May 1990
SUPERSEDING
MIL-L-51501(EA)
14 April 1987

MILITARY SPECIFICATION

LAUNCHERS, GRENADE, SMOKE: SCREENING, RP, M243, M257, AND M259

This specification is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of launchers.

1.2 Classification. Launchers shall be of the following types as specified (see 6.2):

Type	Assembly drawing
M243	13-12-80
M257	13-12-129
M259	13-12-150

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1040

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Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-63150 - Weapons and Support Materiel Standard Quality Assurance Provisions for

STANDARDS

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

13-12-80 - Launcher, Grenade, Smoke: Screening, RP, M243
13-12-90 - Discharger, Smoke Grenade
13-12-128 - Packing and Marking for Discharger, Smoke Grenade
13-12-129 - Launcher, Grenade, Smoke: Screening, RP, M257
13-12-131 - Packing and Marking for Launcher, Grenade, Smoke: Screening, RP, M257
13-12-150 - Launcher, Grenade, Smoke: Screening, RP, M259
13-12-157 - Packing and Marking for Launcher, Grenade, Smoke: Screening, RP, M259
13-12-186 - Packing and Marking for Launcher, Grenade, Smoke: Screening, RP, M243

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P13-12-111 - Arming, Firing Unit
P13-12-165 - Stowage Box, Smoke Grenade

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(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CTF, Rock Island, IL 61229-7300.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 13-12-80 and subsidiary drawings, Drawing 13-12-129 and subsidiary drawings, or Drawing 13-12-150 and subsidiary drawings, as applicable, shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the launcher shall conform to the specifications and drawings listed on Drawing 13-12-80 and subsidiary drawings, Drawing 13-12-129 and subsidiary drawings, or Drawing 13-12-150 and subsidiary drawings, as applicable.

3.2 Discharger leakage. The assembled discharger (Drawing 13-12-90) shall not leak when subjected to an internal air pressure of 1.5 (+0.5 or -0.0) pounds per square inch gage (psig) [10.34 (+3.45 or - 0.0) kilopascals] for a minimum period of 30 seconds at ambient temperature when tested as specified in 4.4.4.1.

3.3 Discharger resistance. The resistance measured between the points shall be in accordance with table I for the discharger when tested as specified in 4.4.4.2.

TABLE I. **Discharger resistance**

From	To	Resistance (ohms)
Receptacle Connector Pin A	Contact pin in barrels	15.0 ± 1.5
Receptacle Connector Pin B	Pin A (receptacle connector)	Infinite resistance
	Pin C (receptacle connector)	Infinite resistance
Receptacle Connector Pin C	Adapter assemblies in all barrels	Zero resistance
	Pin A (receptacle connector)	Infinite resistance

3.4 Insulation resistance. The discharger shall have an insulation resistance of not less than 10 megohms when tested as specified in 4.4.4.3.

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3.5 First article. Prior to the start of regular production, a first article sample of launchers shall be produced (see 6.2) in accordance with this specification for examination and test (see 4.3).

3.6 Workmanship. The assembled launcher shall be free from contamination (oil, rust, grease, or foreign matter) and damage such as dents, cracks, or deformation which could affect its intended use and shall comply with the workmanship requirements of MIL-STD-454 and MIL-W-63150.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the applicable drawing, special packaging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternative inspection provisions. Alternative inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternative procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval. (See 6.3.) When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the quality assurance provisions specified herein. In cases of

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dispute as to whether the contractor's proposed alternative provides equal quality assurance, the provisions of this specification shall apply. All approved alternative inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Sample. The first article sample shall consist of five launchers manufactured and packaged (when applicable) using the same methods, materials, equipment, and processes as will be used during regular production. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.2 Inspections to be performed. As determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests specified in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

4.3.3 Rejection criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. The contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of like components or subassemblies of one interfix number produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.4 and, when specified, table II. Samples shall be selected at random.

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TABLE II. **Sampling**

Lot size	Inspection level					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10
150001 to 500000	1250	346	156	91	40	10
500001 and over	1250	431	187	102	40	10

Numbers under inspection levels indicate sample size; * indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.
Accept the lot represented on zero and reject the lot represented on one or more for all inspection levels.

4.4.3 Inspection procedure. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.4. Failure of any sample item to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.4 Classification of characteristics. Quality conformance examinations and tests shall be as specified in the following classification of characteristics paragraphs. When specified herein, accept on 0 and reject on 1 attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table II for selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.4(a)	Launcher, Grenade, Smoke: Screening, RP, M243 (Dwg 13-12-80), M257 (Dwg 13-12-129), or M259 (Dwg 13-12-150)	SHEET 1 OF 1	As applicable NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Item missing	Table II, level III	3.1.2	VI

NOTES:
VI – Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.4(b)	Discharger, Smoke Grenade		13-12-90
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Circuit resistance	100 percent	3.3	4.4.5.2
Major				
101	Leakage	Table II, level III	3.2	4.4.5.1
102	Insulation resistance	Table II, level III	3.4	4.4.5.3
103	Protective finish missing or incorrect	Table II, level III	3.1	VI
104	Screw type fasteners improperly torqued or loose	Table II, level III	3.1	VI and CE
105	Workmanship	Table II, level III	3.6	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 13-12-128
4.4.4(c)	Packaging and Marking for Discharger, Smoke Grenade		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Item not clean and dry prior to unit packing	Table II, level V	5.1.3	VI
102	Container incorrect	Table II, level V	5.1.3	VI and CE
103	Fiberboard pads missing or incorrect	Table II, level V	5.1.3	VI and CE
104	Sleeve missing or incorrect	Table II, level V	5.1.3	VI and CE
105	Container closure incorrect	Table II, level V	5.1.3	VI and CE
106	Marking incorrect, missing, or illegible	Table II, level V	5.1.3	VI and CE

NOTES:
CE – Commercial inspection equipment
VI – Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE		DRAWING NUMBER
4.4.4(d)	Packaging and Marking for Launcher, Grenade, Smoke: Screening, RP, M257	SHEET 1 OF 1	13-12-131
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Container incorrect	Table II, level V	5.2	VI and CE
102	Container closure incorrect	Table II, level V	5.2	VI and CE
103	Strapping incorrect or missing	Table II, level V	5.2	VI and CE
104	Marking incorrect, missing, or illegible	Table II, level V	5.2	VI and CE

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 13-12-157
4.4.4(e)	Packaging and Marking for Launcher, Grenade, Smoke: Screening, RP, M259		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Container incorrect	Table II, level V	5.2	VI and CE
102	Fiberboard pads incorrect or missing	Table II, level V	5.2	VI and CE
103	Partition incorrect or missing	Table II, level V	5.2	VI and CE
104	Container closure incorrect	Table II, level V	5.2	VI and CE
105	Strapping incorrect or missing	Table II, level V	5.2	VI and CE
106	Marking incorrect, missing, or illegible	Table II, level V	5.2	VI and CE

NOTES:
CE – Commercial inspection equipment
VI – Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 13-12-186
4.4.4(f)	Packaging and Marking for Launcher, Grenade, Smoke: Screening, RP, M243		NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Container incorrect	Table II, level V	5.2	VI and CE
102	Fiberboard pads incorrect or missing	Table II, level V	5.2	VI and CE
103	Void areas not filled as specified	Table II, level V	5.2	VI
104	Container closure incorrect	Table II, level V	5.2	VI and CE
105	Strapping incorrect or missing	Table II, level V	5.2	VI and CE
106	Marking incorrect, missing, or illegible	Table II, level V	5.2	VI and CE

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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4.4.5 Tests.

4.4.5.1 Leakage. The discharger shall be subjected to the specified pressure applied through the test port and submerged in water at ambient temperature to a maximum depth of 3 inches (76 millimeters) below the surface of the water for the required time. A continuous stream of bubbles indicates leakage.

4.4.5.2 Discharger resistance. The resistance shall be measured using an ohmmeter to determine the resistance between the pin (from) and the contact pin or body (to) as specified in 3.3.

4.4.5.3 Insulation resistance. The insulation resistance between pairs of component circuits and between each circuit and ground shall be as specified when measured at 500 volts direct current (vdc) minimum, for a minimum of 30 seconds.

5. PACKAGING

5.1 Preservation, level A.

5.1.1 Stowage box. The stowage box shall be packaged and marked in accordance with SPI P13-12-165.

5.1. Arming/firing unit. The arming/firing unit shall be packaged and marked in accordance with SPI P13-12-111.

5.1.3 Discharger and caps. The discharger and caps shall be packaged and marked as shown on Drawing 13-12-128.

5.2 Packing, level A. Items packaged and marked as specified in 5.1, shall be packed together as shown on Drawing 13-12-157, 13-12-186, or 13-12-131, as applicable (see 6.2).

5.3 Marking. Items packed as specified in 5.2, shall be subsequently marked for storage and shipment in accordance with Drawing 13-12-157, 13-12-186, or 13-12-131, as applicable (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The launchers covered by this specification are intended for use as vehicle mounted launchers for smoke grenades.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Type of launcher required (see 1.2)
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)
(d) First article:
(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.
(e) Applicable packaging drawing (see 5.2 and 5.3)

6.3 Submission of alternative inspection provisions. Propose alternative inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 Elimination of Acceptable Quality Levels and Lot Tolerance Percent Defectives. This specification is certified to be in compliance with current Army Material Command policy for the elimination of Acceptable Quality Levels (AQLs) and Lot Tolerance Percent Defectives (LTPDs) from acquisition documents.

6.5 Subject term (key word) listing.

Arming/firing unit
Discharger
Discharger resistance
Insulation resistance

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1040-A192

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-51501A(EA)	LAUNCHERS, GRENADE, SMOKE: Screening, RP, M243, M257, AND M259

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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